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B - 27

ROCK-PAPER-SCISSORS GAME

ABSTRACT ~

This project is a console-based Rock–Paper–Scissors game developed in C language.

The program allows the user to play against the computer using simple keyboard inputs.

It includes a user-friendly menu with options to play, view rules, check game history, and read about the developer.

The game logic uses random number generation, loops, and conditional statements to determine the winner for each round.

Scores are displayed after each round and saved in a text file for future reference.

This project demonstrates the application of basic programming concepts in a fun and interactive way.

STEPWISE IMPLEMENTATION ~

Step 1: Starting the Program (Main Function)

```
int main()
{
    menu();
    return 0;
}
```

Explanation:

The main() function is the program entry point. It calls the menu() function to start the game interface.

Step 2: Displaying the Main Menu

```
// Displays the main menu
void menu()
{
    int choice;
    do
    {
        printf("\n-----\n");
        printf("          ROCK - PAPER - SCISSORS GAME\n");
        printf("-----\n");
        printf("1. View Rules\n");
        printf("2. Play Game\n");
        printf("3. View History\n");
        printf("4. About Developer\n");
        printf("5. Exit\n");
        printf("-----\n");
        printf("Enter your choice: ");
    } while (choice != 5);
}
```

Explanation:

The menu() function displays a menu with options: View Rules, Play Game, View History, About Developer, and Exit.

It uses a do-while loop to keep showing the menu until the user chooses to exit.

----- ROCK - PAPER - SCISSORS GAME -----

1. View Rules
 2. Play Game
 3. View History
 4. About Developer
 5. Exit
-

Enter your choice:

Step 3: Handling Menu Selection (Switch Statement)

```
switch (choice)
{
    case 1:
        rules();
        break;
    case 2:
        Game();
        break;
    case 3:
        History();
        break;
    case 4:
        Developer();
        break;
    case 5:
        printf("\nThank you for playing!\n");
        break;
```

Explanation:

The user's choice is handled with a switch statement.

Each case calls the corresponding function: rules(), Game(), History(), Developer(), or exits the program.

Step 4: Handling Invalid Menu Input

```
        default:
            printf("\nInvalid choice! Please try again.\n");
    }
```

Explanation:

If the user enters a number outside 1–5, the program prints an error message and prompts again.

This ensures smooth program execution.

```
-----
Enter your choice: 6
```

```
Invalid choice! Please try again.
-----
```

Step 5: Viewing Game Rules

```
// Displays game rules
void rules()
{
    printf("\n===== GAME RULES =====\n");
    printf("1. The game is played between you and the computer.\n");
    printf("2. You must choose one of the following: Rock, Paper, or Scissors.\n");
    printf("3. The computer will also make its choice randomly.\n");
    printf("4. The winner of each round is decided as follows:\n");
    printf("    - Rock beats Scissors (Rock crushes Scissors)\n");
    printf("    - Scissors beats Paper (Scissors cuts Paper)\n");
    printf("    - Paper beats Rock (Paper wraps Rock)\n");
    printf("5. If both you and the computer choose the same option, it's a draw.\n");
    printf("6. You can play multiple rounds; your total score will be displayed.\n");
    printf("7. You can exit anytime by choosing the Exit option from the main menu.\n");
    printf("8. Play smart, have fun, and try to beat the computer!\n");
    printf("=====\n\n");
}
```

Explanation:

The rules() function prints all rules of Rock–Paper–Scissors, explaining how the winner is decided. It guides the player before starting the game.

```
===== GAME RULES =====
1. The game is played between you and the computer.
2. You must choose one of the following: Rock, Paper, or Scissors.
3. The computer will also make its choice randomly.
4. The winner of each round is decided as follows:
    - Rock beats Scissors (Rock crushes Scissors)
    - Scissors beats Paper (Scissors cuts Paper)
    - Paper beats Rock (Paper wraps Rock)
5. If both you and the computer choose the same option, it's a draw.
6. You can play multiple rounds; your total score will be displayed.
7. You can exit anytime by choosing the Exit option from the main menu.
8. Play smart, have fun, and try to beat the computer!
=====
```

Step 6: Starting the Game (Asking Number of Rounds)

```
printf("\nEnter number of rounds: ");
scanf("%d", &rounds);
```

Explanation:

The Game() function asks the player how many rounds they want to play and stores it in a variable.

```
Enter number of rounds: 3
```

Step 7: Generating Computer Choice

```
srand(time(0)); // Seed for random number generation
```

```

int random = rand() % 3; // 0 = Rock, 1 = Paper, 2 = Scissors
if (random == 0)
{
    compChoice = 'r';
}
else if (random == 1)
{
    compChoice = 'p';
}
else
{
    compChoice = 's';
}

```

Explanation:

The computer randomly selects Rock, Paper, or Scissors using `rand() % 3`.

0 = Rock, 1 = Paper, 2 = Scissors. `srand(time(0))` ensures randomness.

Step 8: Taking Player Input

```

printf("\nRound %d\n", i);
printf("Enter your choice (r/p/s): ");
scanf(" %c", &playerChoice);

```

Explanation:

The player enters their choice (r, p, or s) using `scanf()`.

Input is read carefully to avoid issues with leftover newline characters.

```

Round 1
Enter your choice (r/p/s): r

```

Step 9: Displaying Round Choices

```

printf("You chose: %c, Computer chose: %c\n", playerChoice, compChoice);

```

Explanation:

Both the player's and computer's choices are displayed for clarity before determining the winner.

You chose: r, Computer chose: p

Step 10: Determining Winner for Each Round

```
// Determine winner
if (playerChoice == compChoice)
    printf("It's a draw!\n");
else
    Loading...
    ((playerChoice == 'r' && compChoice == 's') ||
     (playerChoice == 'p' && compChoice == 'r') ||
     (playerChoice == 's' && compChoice == 'p'))
    {
        printf("You win this round!\n");
        playerScore++;
    }
else
    {
        printf("Computer wins this round!\n");
        compScore++;
    }
}
```

Explanation:

The program uses if-else conditions to compare choices:

Same choice → Draw

Player wins → Player score increments

Else → Computer wins → Computer score increments

Computer wins this round!

Step 11: Displaying Final Scores


```
// Final result
printf("\n-----\n");
printf("Final Scores:\n");
printf("You: %d\nComputer: %d\n", playerScore, compScore);

if (playerScore > compScore)
    printf("\nCongratulations! You are the overall winner!\n");
else if (playerScore < compScore)
    printf("\nComputer wins the game! Better luck next time!\n");
else
    printf("\nGame ended in a draw!\n");
printf("-----\n");
```

Explanation:

After all rounds, total scores are printed along with the overall winner or a draw message.

```
-----
Final Scores:
You: 2
Computer: 1

Congratulations! You are the overall winner!
-----
```

Step 12: Saving Game Results to File

```
// File handling part - save result
FILE *fp;
fp = fopen("game_history.txt", "a");
if (fp != NULL)
{
    fprintf(fp, "Total Rounds: %d | You: %d | Computer: %d | ", rounds, playerScore, compScore);
    if (playerScore > compScore)
        fprintf(fp, "Result: You Won\n");
    else if (playerScore < compScore)
        fprintf(fp, "Result: Computer Won\n");
    else
        fprintf(fp, "Result: Draw\n");
    fclose(fp);
}
```

Explanation:

The program appends results to game_history.txt using fopen(), fprintf(), and fclose().

This keeps a record of rounds, scores, and game outcomes.

Step 13: Replay Option

```
// Replay option
char playAgain;
printf("\nDo you want to play again? (y/n): ");
scanf(" %c", &playAgain);
if (playAgain == 'y' || playAgain == 'Y')
    Game();
}
```

Explanation:

At the end of the game, the player is asked if they want to play again.

If yes, the Game() function is called recursively.

```
Do you want to play again? (y/n): y
```

Step 14: Viewing Game History

```
// View History function
void History()
{
    FILE *fp;
    char line[200];

    fp = fopen("game_history.txt", "r");
    if (fp == NULL)
    {
        printf("\nNo game history found yet.\n");
        return;
    }

    printf("\n===== GAME HISTORY =====\n");
    while (fgets(line, sizeof(line), fp))
    {
        printf("%s", line);
    }
    fclose(fp);
    printf("=====\n");
}
```

Explanation:

The History() function reads and prints all previous results from game_history.txt.

If no history exists, it prints a “No game history found” message.

If no history exist --

```
-----
Enter your choice: 3

No game history found yet.

-----
```

If history exist –

```
===== GAME HISTORY =====
Total Rounds: 3 | You: 0 | Computer: 2 | Result: Computer Won
Total Rounds: 3 | You: 2 | Computer: 1 | Result: You Won
Total Rounds: 3 | You: 0 | Computer: 3 | Result: Computer Won
Total Rounds: 3 | You: 2 | Computer: 1 | Result: You Won
Total Rounds: 1 | You: 0 | Computer: 1 | Result: Computer Won
Total Rounds: 2 | You: 0 | Computer: 1 | Result: Computer Won
Total Rounds: 2 | You: 0 | Computer: 0 | Result: Draw
Total Rounds: 5 | You: 2 | Computer: 0 | Result: You Won
=====
```

Step 15: About Developer

```
// About developer section
void Developer()
{
    printf("\n-----\n");
    printf("                ABOUT DEVELOPER\n");
    printf("-----\n");
    printf("Name           : Sanchita Luthra\n");
    printf("Course          : B.Tech\n");
    printf("Semester        : 1st Semester\n");
    printf("University      : UPES\n");
    printf("Project Title   : Rock-Paper-Scissors Game\n");
    printf("Language Used   : C\n");
    printf("Date            : November 2025\n");
    printf("-----\n");
    printf("Thank you for viewing my project!\n");
    printf("-----\n");
}
```

Explanation:

The Developer() function displays the developer's name, course, university, project title, and language used.

```
-----
                ABOUT DEVELOPER
-----
Name           : Sanchita Luthra
Course          : B.Tech
Semester        : 1st Semester
University      : UPES
Project Title   : Rock-Paper-Scissors Game
Language Used   : C
Date            : November 2025
-----
Thank you for viewing my project!
-----
```

Step 16: Returning to Menu After Actions

```
do
{
    printf("\n-----\n");
    printf("          ROCK - PAPER - SCISSORS GAME\n");
    printf("-----\n");
    printf("1. View Rules\n");
    printf("2. Play Game\n");
    printf("3. View History\n");
    printf("4. About Developer\n");
    printf("5. Exit\n");
    printf("-----\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            rules();
            break;
        case 2:
            Game();
            break;
        case 3:
            History();
            break;
        case 4:
            Developer();
            break;
        case 5:
            printf("\nThank you for playing!\n");
            break;
        default:
            printf("\nInvalid choice! Please try again.\n");
    }
} while (choice != 5);
```

Explanation:

After viewing rules, history, or developer info, the program returns to the main menu automatically.

This ensures seamless navigation.

ROCK - PAPER - SCISSORS GAME

1. View Rules
 2. Play Game
 3. View History
 4. About Developer
 5. Exit
-

Enter your choice:

COMPLETE SOURCE CODE

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

// Function prototypes
void menu();
void rules();
void Game();
void Developer();
void History();

int main()
{
    menu();
    return 0;
}

// Displays the main menu
void menu()
{
    int choice;
    do
    {
        printf("\n-----\n");
        printf("          ROCK - PAPER - SCISSORS GAME\n");
        printf("-----\n");
        printf("1. View Rules\n");
        printf("2. Play Game\n");
        printf("3. View History\n");
        printf("4. About Developer\n");
        printf("5. Exit\n");
        printf("-----\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                rules();
                break;
            case 2:
                Game();
                break;
            case 3:
                History();
                break;
            case 4:
                Developer();
                break;
            case 5:
```

```

        printf("\nThank you for playing!\n");
        break;
    default:
        printf("\nInvalid choice! Please try again.\n");
    }
} while (choice != 5);
}

// Displays game rules
void rules()
{
    printf("\n===== GAME RULES =====\n");
    printf("1. The game is played between you and the computer.\n");
    printf("2. You must choose one of the following: Rock, Paper, or Scissors.\n");
    printf("3. The computer will also make its choice randomly.\n");
    printf("4. The winner of each round is decided as follows:\n");
    printf("    - Rock beats Scissors (Rock crushes Scissors)\n");
    printf("    - Scissors beats Paper (Scissors cuts Paper)\n");
    printf("    - Paper beats Rock (Paper wraps Rock)\n");
    printf("5. If both you and the computer choose the same option, it's a draw.\n");
    printf("6. You can play multiple rounds; your total score will be displayed.\n");
    printf("7. You can exit anytime by choosing the Exit option from the main menu.\n");
    printf("8. Play smart, have fun, and try to beat the computer!\n");
    printf("=====\n\n");
}

// Handles the main game logic
void Game()
{
    int rounds, i;
    char playerChoice, compChoice;
    int playerScore = 0, compScore = 0;

    srand(time(0)); // Seed for random number generation

    printf("\nEnter number of rounds: ");
    scanf("%d", &rounds);

    for (i = 1; i <= rounds; i++)
    {
        int random = rand() % 3; // 0 = Rock, 1 = Paper, 2 = Scissors
        if (random == 0)
        {
            compChoice = 'r';
        }
        else if (random == 1)
        {
            compChoice = 'p';
        }
        else
        {
            compChoice = 's';
        }
    }
}

```



```

printf("\nRound %d\n", i);
printf("Enter your choice (r/p/s): ");
scanf(" %c", &playerChoice);

printf("You chose: %c, Computer chose: %c\n", playerChoice, compChoice);

// Determine winner
if (playerChoice == compChoice)
    printf("It's a draw!\n");
else if
    ((playerChoice == 'r' && compChoice == 's') ||
     (playerChoice == 'p' && compChoice == 'r') ||
     (playerChoice == 's' && compChoice == 'p'))
{
    printf("You win this round!\n");
    playerScore++;
}
else
{
    printf("Computer wins this round!\n");
    compScore++;
}
}

// Final result
printf("\n-----\n");
printf("Final Scores:\n");
printf("You: %d\nComputer: %d\n", playerScore, compScore);

if (playerScore > compScore)
    printf("\nCongratulations! You are the overall winner!\n");
else if (playerScore < compScore)
    printf("\nComputer wins the game! Better luck next time!\n");
else
    printf("\nGame ended in a draw!\n");
printf("-----\n");

// File handling part - save result
FILE *fp;
fp = fopen("game_history.txt", "a");
if (fp != NULL)
{
    fprintf(fp, "Total Rounds: %d | You: %d | Computer: %d | ", rounds, playerScore,
compScore);
    if (playerScore > compScore)
        fprintf(fp, "Result: You Won\n");
    else if (playerScore < compScore)
        fprintf(fp, "Result: Computer Won\n");
    else
        fprintf(fp, "Result: Draw\n");
    fclose(fp);
}

// Replay option

```

```

char playAgain;
printf("\nDo you want to play again? (y/n): ");
scanf(" %c", &playAgain);
if (playAgain == 'y' || playAgain == 'Y')
    Game();
}

// View History function
void History()
{
    FILE *fp;
    char line[200];

    fp = fopen("game_history.txt", "r");
    if (fp == NULL)
    {
        printf("\nNo game history found yet.\n");
        return;
    }

    printf("\n===== GAME HISTORY =====\n");
    while (fgets(line, sizeof(line), fp))
    {
        printf("%s", line);
    }
    fclose(fp);
    printf("=====\n");
}

// About developer section
void Developer()
{
    printf("\n-----\n");
    printf("          ABOUT DEVELOPER\n");
    printf("-----\n");
    printf("Name       : Sanchita Luthra\n");
    printf("Course      : B.Tech\n");
    printf("Semester    : 1st Semester\n");
    printf("University  : UPES\n");
    printf("Project Title : Rock-Paper-Scissors Game\n");
    printf("Language Used : C\n");
    printf("Date        : November 2025\n");
    printf("-----\n");
    printf("Thank you for viewing my project!\n");
    printf("-----\n");
}

```

TEST CASES

<u>Test Case ID</u>	<u>Player Input</u>	<u>Possible Computer Input</u>	<u>Result / Notice</u>
TC1	r	r	Draw
		p	Computer wins
		s	Player wins
TC2	p	r	Player wins
		p	Draw
		s	Computer wins
TC3	s	r	Computer wins
		p	Player wins
		s	Draw
TC4	Multiple Rounds (3)	Round 1 : r/p/s Round 2 : r/p/s Round 3 : r/p/s	Each round follows logic rock beats scissors , paper beats rock , scissors beat paper Final score = sum of round results
TC5	Invalid Input (e.g., x)	r/p/s	Treated as wrong input ~ computer wins round (as code currently counts invalid input so computer wins)

TC1~

```
Enter number of rounds: 1
```

```
Round 1
```

```
Enter your choice (r/p/s): r
```

```
You chose: r, Computer chose: s
```

```
You win this round!
```

```
-----  
Final Scores:
```

```
You: 1
```

```
Computer: 0
```

```
Congratulations! You are the overall winner!
```

TC2~

```
Round 1
```

```
Enter your choice (r/p/s): p
```

```
You chose: p, Computer chose: s
```

```
Computer wins this round!
```

```
-----  
Final Scores:
```

```
You: 0
```

```
Computer: 1
```

```
Computer wins the game! Better luck next time!
```

TC3~

```
Round 1
Enter your choice (r/p/s): s
You chose: s, Computer chose: s
It's a draw!
```

```
-----
Final Scores:
```

```
You: 0
```

```
Computer: 0
```

```
Game ended in a draw!
-----
```

TC4~

```
Enter number of rounds: 3
```

```
Round 1
```

```
Enter your choice (r/p/s): r
```

```
You chose: r, Computer chose: s
```

```
You win this round!
```

```
Round 2
```

```
Enter your choice (r/p/s): p
```

```
You chose: p, Computer chose: r
```

```
You win this round!
```

```
Round 3
```

```
Enter your choice (r/p/s): s
```

```
You chose: s, Computer chose: s
```

```
It's a draw!
```

```
-----
Final Scores:
```

```
You: 2
```

```
Computer: 0
```

```
Congratulations! You are the overall winner!
```

```
-----
Do you want to play again? (y/n): █
```

TC5 ~

```
Enter number of rounds: 1
```

```
Round 1
```

```
Enter your choice (r/p/s): x
```

```
You chose: x, Computer chose: r
```

```
Computer wins this round!
```

```
-----  
Final Scores:
```

```
You: 0
```

```
Computer: 1
```

```
Computer wins the game! Better luck next time!
```

```
-----
```


